KNN Regression

In [ ]:
```
K-Nearest Neighbours is a supervidsed Machine learning algorithn
it can be used for
1)Classification   --->Predicts category
2)Regression   ----->predicts a numerical value


To predict a new value ,look at the K most similar/nearest data points and use
their values to make the prediction

For KNN Regression we generally take the average of target values of the K nearest
Neighbours


Similar
Related


Area          Price

500           30lakhs
600           35
700           40
800           45
900           50

750  --->?

if k=3

Nearest houses are:

700 sqrt
800
600

predict:

700+600+800/3=
```

In [4]:
```
700+600+800
```

Out[4]:
```
2100
```

In [8]:
```
2100/3
```

Out[8]:
```
700.0
```

In [ ]:
```
Step1: Take a training data

X=Features
Y=target

Area --->Price


step 2: select K

K=p1,p2,p3 ---->Pn



K tells us how many neighbours we should consider

Step 3:Calculate Distance

For single feature:

Distance=|X1-X2|

for multiple features:

Euclidean Distance(d)=
d1
d2
d3

Step4-->Sort the distances from smallest to largest

Step-5 Take average

y_predict=d1+d2+d3/3


Lazy learner:

y=mx+c

KNN remembers the training data
and calculate neighbours when a new prediction is requested

100
010

standerdscaler=x-mean/sd

10,110


Age       salary

25        50000
30        60000

Note:

KNN is Distance based algorithm so feature scalling
is very important

Step 6

Choosing K:

K=1

Small K:
 Advantage: ---More sensitive to local patterns
Disadvantage
Can be affected by noise
May overfit

Large K:

Smooth prediction
less sensitive to noise

Can Underfit

K small---->Overfit
K Large  --->Underfit

CV  --->Cross validation
```

In [10]:
```
data={
    'Area':[200,600,700,800,900,1000,1100,1200,1300,1400],
    'Bedrooms':[1,1,2,2,2,3,3,3,4,4],
    'Price':[30,35,40,45,50,55,60,65,72,80]
}
```

In [12]:
```
data
```

Out[12]:
```
{'Area': [200, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400],
 'Bedrooms': [1, 1, 2, 2, 2, 3, 3, 3, 4, 4],
 'Price': [30, 35, 40, 45, 50, 55, 60, 65, 72, 80]}
```

In [16]:
```
import pandas as pd
df=pd.DataFrame(data)
```

In [18]:
```
df
```

Out[18]:

	Area	Bedrooms	Price
0	200	1	30
1	600	1	35
2	700	2	40
3	800	2	45
4	900	2	50
5	1000	3	55
6	1100	3	60
7	1200	3	65
8	1300	4	72
9	1400	4	80

In [20]:
```
Y=df[['Price']]
X=df.drop(labels=['Price'],axis=1)
```

In [22]:
```
Y
```

Out[22]:

	Price
0	30
1	35
2	40
3	45
4	50
5	55
6	60
7	65
8	72
9	80

In [24]:
```
X
```

Out[24]:

	Area	Bedrooms
0	200	1
1	600	1
2	700	2
3	800	2
4	900	2
5	1000	3
6	1100	3
7	1200	3
8	1300	4
9	1400	4

In [ ]:

Train Test split

In [27]:
```
from sklearn.model_selection import train_test_split
xtrain,xtest,ytrain,ytest=train_test_split(X,Y,test_size=0.2,random_state=42)
```

Feature scaling

In [30]:
```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()

Xtrain_scaled=sc.fit_transform(xtrain)
Xtest_scaled=sc.fit_transform(xtest)
```

In [34]:
```
Xtrain_scaled
```

Out[34]:
```
array([[ 0.25680509,  0.57735027],
       [-2.09112719, -1.73205081],
       [ 0.84378816,  0.57735027],
       [-0.62366951, -0.57735027],
       [ 1.43077123,  1.73205081],
       [-0.03668644, -0.57735027],
       [-0.33017798, -0.57735027],
       [ 0.55029663,  0.57735027]])
```

In [36]:
```
Xtest_scaled
```

Out[36]:
```
array([[ 1.,  1.],
       [-1., -1.]])
```

Model

In [39]:
```
from sklearn.neighbors import KNeighborsRegressor
model=KNeighborsRegressor(n_neighbors=3)
```

Train the model

In [42]:
```
model.fit(Xtrain_scaled,ytrain)
```

Out[42]:
```
▾      KNeighborsRegressor
KNeighborsRegressor(n_neighbors=3)
```

Make Prediction

In [45]:
```
y_pred=model.predict(Xtest_scaled)
```

In [47]:
```
y_pred
```

Out[47]:
```
array([[68.33333333],
       [45.        ]])
```

Evalute the Model

In [52]:
```
from sklearn.metrics import mean_absolute_error,mean_squared_error
```

In [54]:
```
mae=mean_absolute_error(ytest,y_pred)
print("MAE: ",mae)
```

```
MAE:  6.833333333333336
```

In [ ]:
```
Area=750
Bedroom=2
```

In [68]:
```
new_house=[[3000,5]]

new_house_scaled=sc.transform(new_house)
#print(new_house_scaled)

print(model.predict(new_house_scaled))
```

```
[[68.33333333]]
```

In [ ]:
```
x-mean/sd

x
2   -->0.3
3
4
5
xmean=3

(-1)^2/3
```

In [70]:
```
1/3
```

Out[70]:
```
0.3333333333333333
```

In [78]:
```
xtrain.mean()
```

Out[78]:
```
Area        912.5
Bedrooms      2.5
dtype: float64
```

In [80]:
```
xtrain.std()
```

Out[80]:
```
Area        364.250699
Bedrooms      0.925820
dtype: float64
```

In [82]:
```
xtest.mean()
```

Out[82]:
```
Area        950.0
Bedrooms      2.5
dtype: float64
```

In [84]:
```
xtest.std()
```

Out[84]:
```
Area        494.974747
Bedrooms      2.121320
dtype: float64
```

In [ ]: